CS 532, Spring 2019: Quiz 4

NAME:

Problem 1. Consider a point $\mathbf{x}_A = (x, y, w)^\top$ in image A, and a Fundamental matrix of the form

$$\mathbf{F}_{AB} = \begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_1 \\ \mathbf{f}_2 \\ \mathbf{f}_3 \end{bmatrix}$$

What is the expression of the epipolar line corresponding to $\mathbf{x}_A$ in image B. Make a sketch.

Problem 2. Assuming the same Fundamental matrix as in Problem 1, write the explicit equation for the epipolar constraint $\mathbf{x}_B^\top \mathbf{F}_{AB} \mathbf{x}_A$ for $\mathbf{x}_A = (1, 0, 1)^\top$ and $\mathbf{x}_B = (0, 1, 1)^\top$. Rewrite the expression in terms of vector products.